

Elastic Observability AI Agents Behind the Scenes





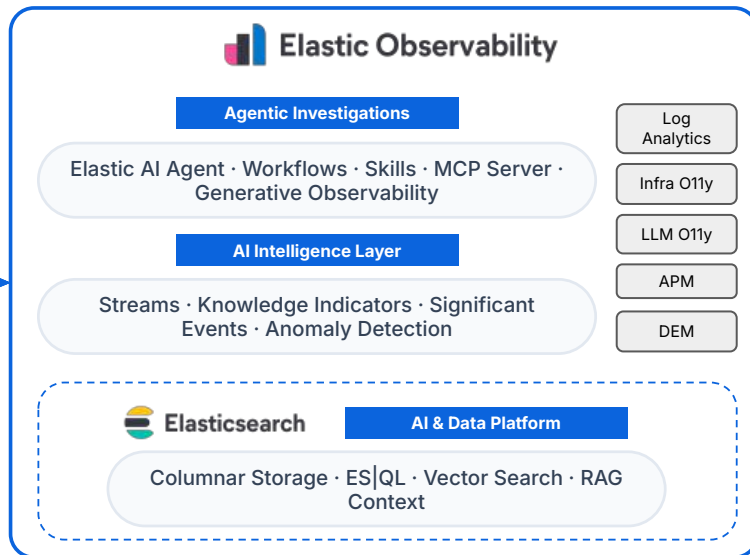
Josh Phifer

Principal Solutions Architect,
Global Public Sector

Elastic Agentic Observability Platform for the AI era.

All Signals

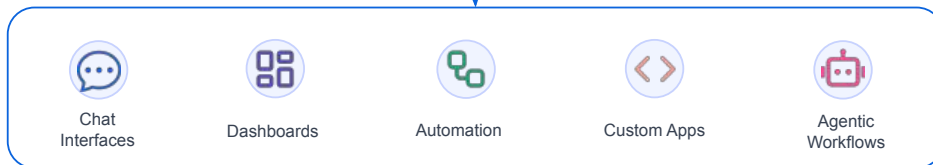
- Logs
- Metrics (Prometheus + OTel)
- Traces
- Synthetics
- Profiles
- Security
- More



Insights to Action

- Anomalies Detected
- Root Cause Identified
- Guided Investigations
- Actions/ Remediations Executed

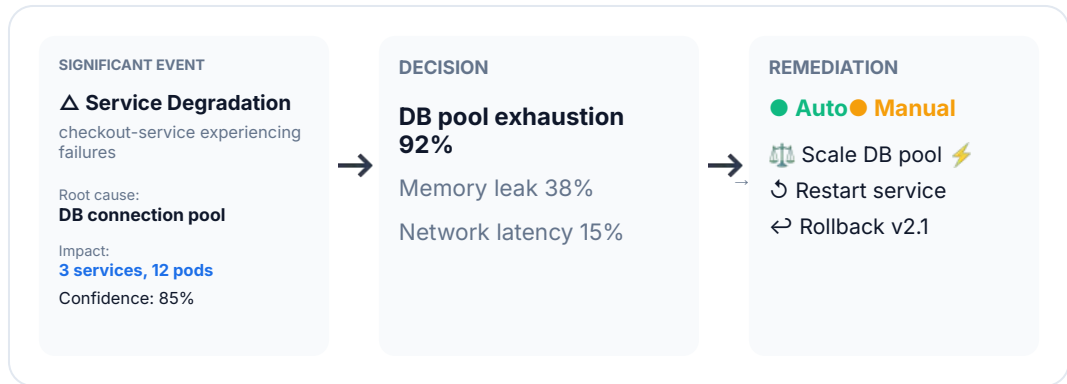
MCP Apps & Reusable Skills



Claude
Cursor
etc

The Agent Recommends. You Decide. Done.

Agents use the live system model to determine root cause, **rank remediation options by confidence**, and execute — **autonomously** or **with a human in the loop**.



⚡ **Autonomous Execution**
Direct action on high confidence

OR

👤 **Human Validation**
Operator review and approval

```
elastic-agent - production cluster

$ elastic --namespace production
> Connecting to cluster...
> Fetching Significant Events...

⚠ INCIDENT · HIGH — payment-service
cause: Redis OOM · error_rate ↑94% · p99 ↑8.4s
blast radius: payment → checkout → orders
confidence: 85% · 3 services · 12 pods

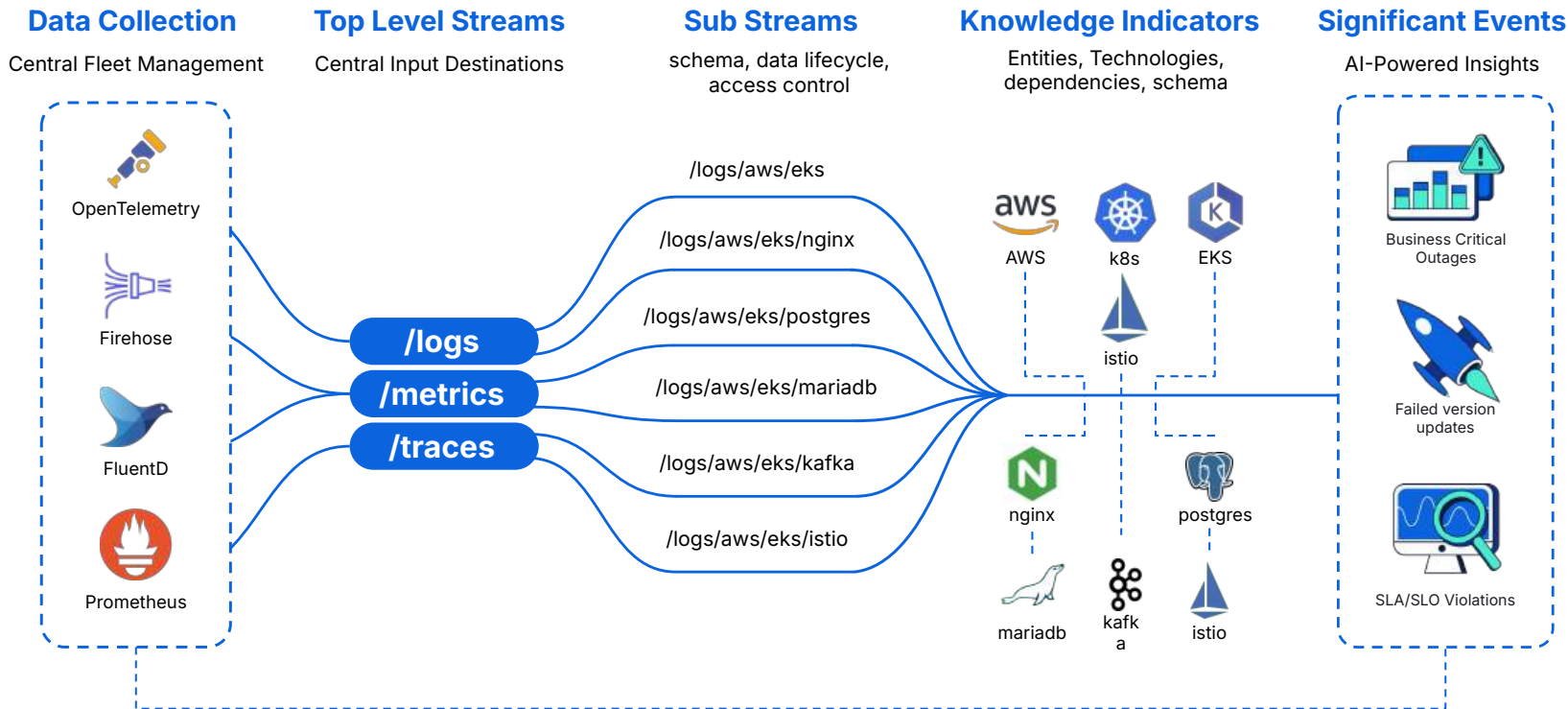
> Root cause: DB connection pool exhaustion
> Proposing remediation plan...
> run remediation (auto)

kubectl rollout restart ✓ (23s)
redis CONFIG SET maxmemory ✓ (1s)
helm pool.max=200 ✓ (2s)

✓ Resolved MTTD 47s MTTR 2m 11s
```

A system that understands, decides, acts, and continuously adapts.

Stream Anything, and get instant insights



Generative AI extracts meaning from raw signals.

Now with native **Prometheus** support in Elasticsearch

Prometheus is the standard. PromQL is the language. **Elastic removes all friction.**

Prometheus-native Ingest & PromQL

"Native Prometheus" with native ingest. PromQL runs natively in Kibana.

5x-25x Faster

vs. Prometheus & Mimir

Tech Preview

Storage Efficiency Gain

Store more metrics at lower cost with massive foundation improvements.

2.5x+ better

vs Prometheus

GA

Migration Tooling

Automatically migrate dashboards and alerting.

1 DAY

to fully operational

Tech Preview

Grafana users: point at Elastic as a backend — same queries, same dashboards, zero retraining.



SREs already speak PromQL.

Now Kibana does too.

80%

of queries in the top 100
dashboards

natively executed in Kibana



Native in Kibana

No plugins, no adapters.
Runs wherever ES|QL is
supported.

Rich Analytics

rate, sum by, max_over_time
— works out-of-the-box.

25× Faster

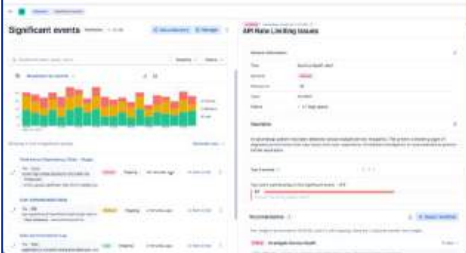
Same queries as Prometheus
with significantly better
performance.

Agentic First Experience

Visualize, analyze, investigate with [context-rich experiences in the platform of your choice](#)

UI in Kibana

- Significant Events landing page across all streams
- Full backtracking: Sig Event → Events → Rules → KIs
- Chat for human-in-the-loop investigation & remediation



Agent Builder

- Chat with any pipeline stage via tools
- Significant Events RCA skill loaded automatically
- Server-side Kibana Workflows — visible and debuggable



CLI / Skills

```
$ claude --namespace production  
  
> You: check for incidents  
> Fetching Significant Events...  
  
✗ INCIDENT · HIGH — payment-service  
  cause: Redis OOM · error_rate ↑94% · p99 ↑8.4s  
  blast radius: payment → checkout → orders  
  
> You: run the remediation  
> Executing plan (confidence 0.87)...  
  kubectl rollout restart ✓ (23s)  
  redis CONFIG SET noeviction ✓ (1s)  
  helm pool.max=200 ✓ (2s)  
  
✓ Resolved MTTD 47s MTTR 2m 11s
```

Use your own Agentic Workflows

Get results in plain language, inside your existing Agents and workflows.



Native MCP server

Connect any Claude-compatible client directly to your Elastic cluster.



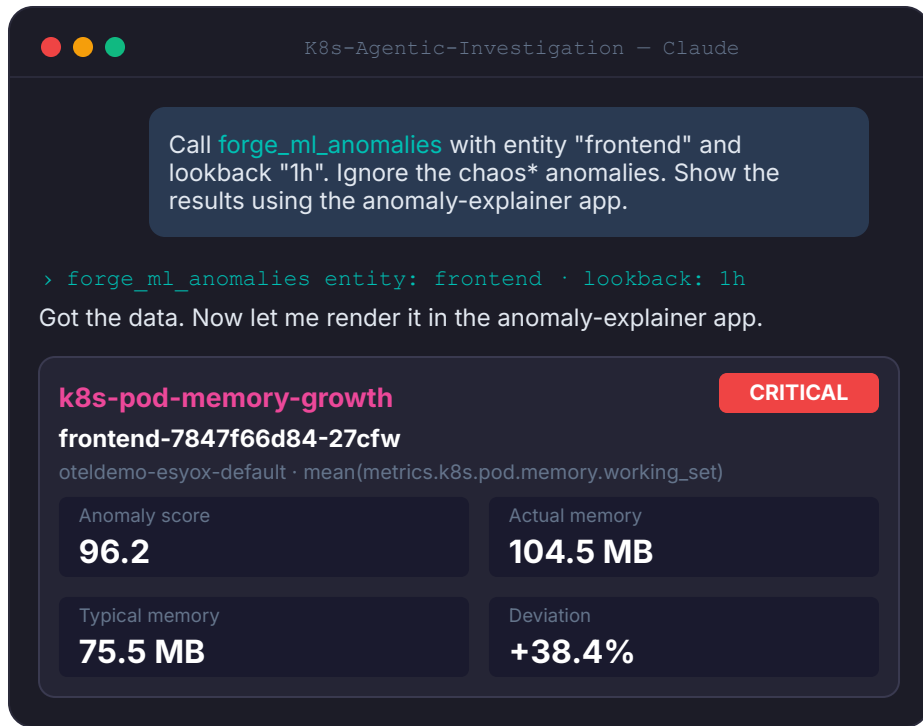
Skills loaded automatically

Significant Events RCA, anomaly detection, and remediation — always available.



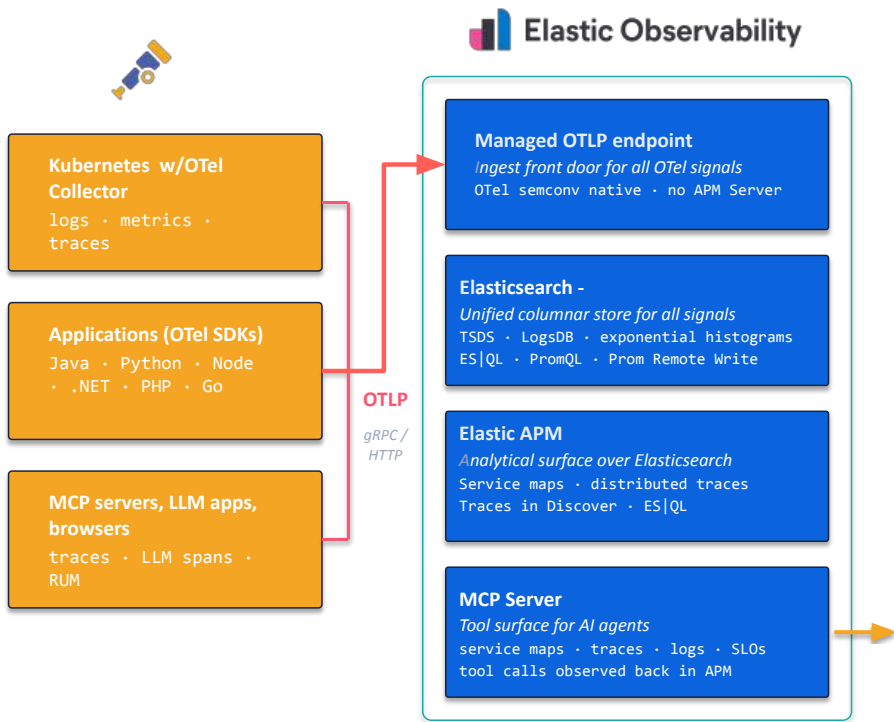
Surface-aware rendering

Results rendered as rich cards, not raw JSON — built for the interface you're using.



Investigate Apps with OpenTelemetry, APM and Agentic skills

Any OTEL signal ingested via OTLP. Stored in Elasticsearch. Analyze with APM and your own agents



Use Significant Events, APM, with Elastic Agent

```
claude - sre@incident-1742

❑ Loaded Elastic Agent Skills: service-health · logs-search · manage-slos · llm-obs

> Find out why transaction 01ba6cf8...3278a1 is having issues over the last 24 hours.

• Using skill: service-health
• Calling tool: apm.get_service_map
• Calling tool: apm.get_trace(01ba6cf8...3278a1)

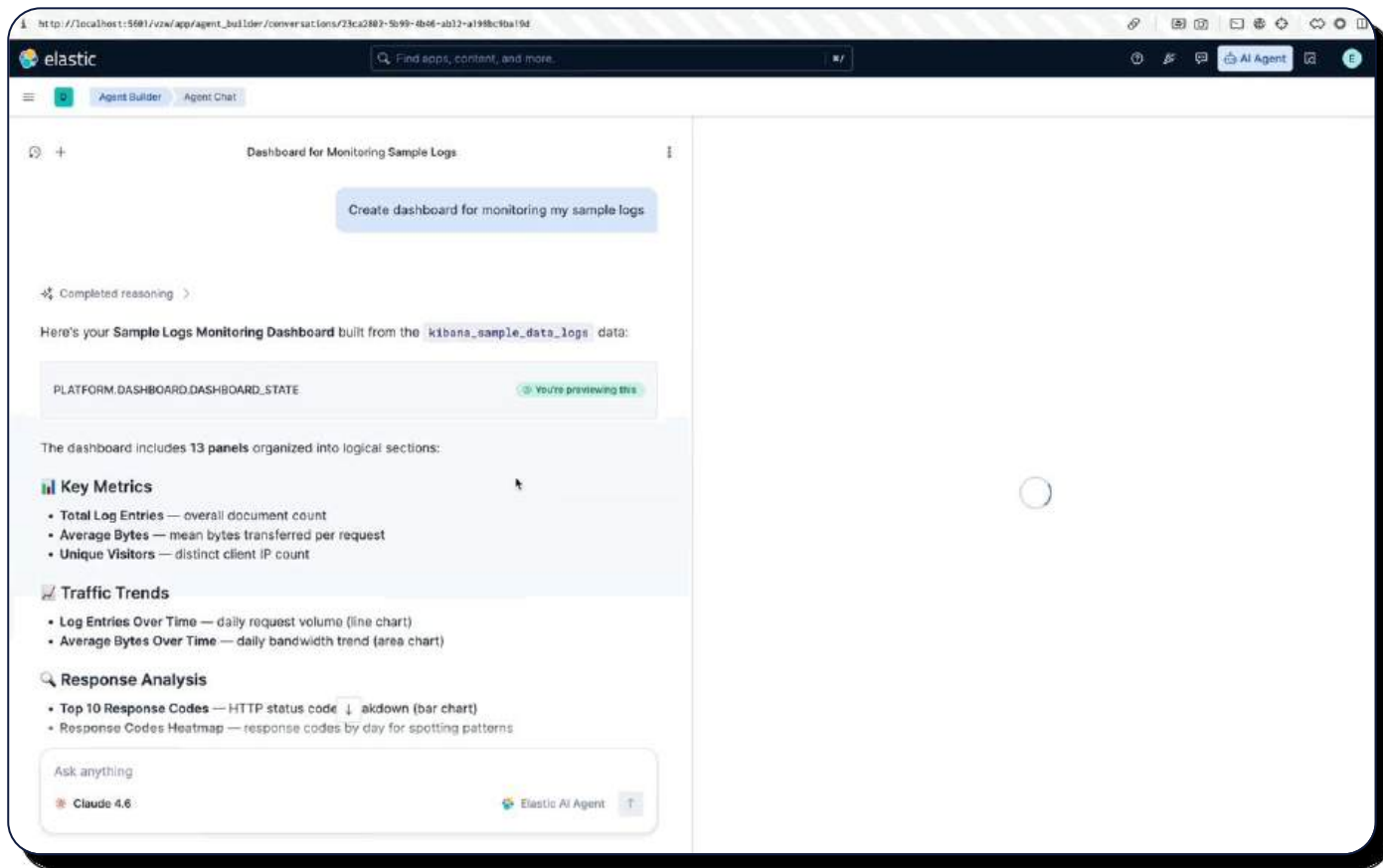
service map from Elastic APM (via MCP)
graph LR
    frontend[frontend] --> cart[cart]
    cart -- "503 / 4.2s" --> payment[payment]
    payment --> db[db]
```

Root cause: payment-service exceeded connection pool to db. 73% of failed spans hit `ConnectionTimeoutException` in last 24h. Started at 14:22 UTC, correlates with deploy `v2.14.1` (pool size 10→5).

Recommend: rollback or restore pool size to 10.

Investigate with Elastic Agent Skills in your own Agent.

Intuitive visualizations with Generative Observability



Elastic recognized as a **Leader** in the 2025 Gartner® Magic Quadrant™ for Observability Platforms.

Gartner, Magic Quadrant for Observability Platforms,
Gregg Siegfried, Matt Crossley, Padraig Byrne, Andre Bridges, Martin Caren 7 July 2025.

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Demo!



Thank you!

